

AWS EBS with Terraform

To Begin with the Lab

Summary of the Lab

This lab demonstrated how to provision and manage an AWS EBS volume using Terraform. After creating Terraform configuration files, initializing Terraform, and applying the configuration, an EC2 instance and EBS volume were created. The EBS volume was formatted, mounted to /data, and verified. Since the mount was lost after reboot, a persistent entry was added to /etc/fstab, ensuring the EBS volume automatically remounts and remains available across instance reboots.

Prerequisite for the lab

Running EC2 instance with installed terraform

Now create a new directory and Navigate to the directory.

Create three files

Createinstance.tf

Variable.tf

Provider.tf

Then the run the command

```
terraform init
```

```
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume# terraform init
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.50.0...
- Installed hashicorp/aws v6.50.0 (signed by HashiCorp)

Initializing the backend...

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume#
```

Now run the command

```
ssh-keygen -f levelup_key
```

Hit enter

```

root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume# ssh-keygen -f levelup_key
Generating public/private ed25519 key pair.
Enter passphrase for "levelup_key" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in levelup_key
Your public key has been saved in levelup_key.pub
The key fingerprint is:
SHA256:+VLLMUwcVapL6RNSf/oBFyX+6pYjbJMcuDOPDXF0F0w root@ip-172-31-17-1
The key's randomart image is:
+--[ED25519 256]--+
|
| .oo=Eo|
| o.. +o|
| B +..|
| . = B o.|
| S +.B...o|
| o*.oo+|
| . o++ooo|
| .++Bo+..|
| .+o+...|
+-----[SHA256]-----+
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume#

```

Now run the command

```
terraform plan
```

it will ask you to enter the secret key

then it will show that terraform will add 4 resources

```

# aws_volume_attachment.ebs-volume-1-attachment will be created
+ resource "aws_volume_attachment" "ebs-volume-1-attachment" {
+   device_name = "/dev/xvdx"
+   id          = (known after apply)
+   instance_id = (known after apply)
+   region      = "us-east-1"
+   volume_id   = (known after apply)
+ }
Plan: 4 to add, 0 to change, 0 to destroy.

```

Now run the command

```
terraform apply
```

then it will prompt you to enter the value

```

Plan: 4 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

```

type: yes

and press enter

it will start creating the following resource

```

aws_key_pair.levelup_key: Creating...
aws_ebs_volume.ebs-volume-1: Creating...
aws_key_pair.levelup_key: Creation complete after 0s [id=levelup_key]
aws_instance.MyFirstInstnace: Creating...

```

Resources successfully created

```

aws_key_pair.levelup_key: Creating...
aws_ebs_volume.ebs-volume-1: Creating...
aws_key_pair.levelup_key: Creation complete after 1s [id=levelup_key]
aws_instance.MyFirstInstnace: Creating...
aws_ebs_volume.ebs-volume-1: Still creating... [00m10s elapsed]
aws_instance.MyFirstInstnace: Still creating... [00m10s elapsed]
aws_ebs_volume.ebs-volume-1: Creation complete after 11s [id=vol-09df25eba464bf469]
aws_instance.MyFirstInstnace: Still creating... [00m20s elapsed]
aws_instance.MyFirstInstnace: Still creating... [00m30s elapsed]
aws_instance.MyFirstInstnace: Creation complete after 32s [id=i-0007ffa4dfa50b034]
aws_volume_attachment.ebs-volume-1-attachment: Creating...
aws_volume_attachment.ebs-volume-1-attachment: Still creating... [00m10s elapsed]
aws_volume_attachment.ebs-volume-1-attachment: Still creating... [00m20s elapsed]
aws_volume_attachment.ebs-volume-1-attachment: Creation complete after 20s [id=vai-3052517475]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud/EBS_Volume#

```

Instances (1/2) info Last updated less than a minute ago Connect Instance state Actions Launch instances

Saved filter sets: Running

Instance state = running Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	terraform	i-0bbe29e1171b02360	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-54-227-124-69.co.
☒	custom_instance	i-0007ffa4dfa50b034	Running	t2.micro	2/2 checks passed	us-east-1a	ec2-98-88-36-117.com

i-0007ffa4dfa50b034 (custom_instance)

Filter block devices

	Volume ID	Device name	Volume size (GiB)	Volume State	Attachment status	EBS card index	Attachment time
☒	vol-00d6eb43666e9ff6b	/dev/sda1	8	In-use	Attached	0	2026/06/17 22
☐	vol-09df25eba464bf469	/dev/xvdx	10	In-use	Attached	0	2026/06/17 22

Check existing disks

```
df -h
```

Create a filesystem on the EBS volume

The EBS volume was attached as /dev/xvdx.

```
mkfs.ext4 /dev/xvdx
```

This formats the disk with the ext4 filesystem.

Create a mount point

```
mkdir -p /data
```

Mount the EBS volume

```
mount /dev/xvdx /data
```

Verify the mount

```
df -h
```

You should see the 10 GB EBS volume mounted on /data.

```

root@ip-172-31-30-211:~# mkfs.ext4 /dev/xvdh
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 2621440 4k blocks and 655360 inodes
Filesystem UUID: 22c78442-0c27-49c1-a1cc-5c1baled36bf
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@ip-172-31-30-211:~# mkdir -p /data
root@ip-172-31-30-211:~# mount /dev/xvdh /data
root@ip-172-31-30-211:~# df -h

```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/root	6.7G	2.1G	4.6G	31%	/
tmpfs	476M	0	476M	0%	/dev/shm
tmpfs	191M	884K	190M	1%	/run
tmpfs	476M	0	476M	0%	/tmp
none	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
none	1.0M	0	1.0M	0%	/run/credentials/systemd-resolved.service
/dev/xvda13	989M	96M	827M	11%	/boot
/dev/xvda15	105M	6.3M	99M	7%	/boot/efi
none	1.0M	0	1.0M	0%	/run/credentials/systemd-networkd.service
none	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
none	1.0M	0	1.0M	0%	/run/credentials/serial-getty@ttyS0.service
tmpfs	96M	8.0K	96M	1%	/run/user/1000
/dev/xvdh	9.8G	2.1M	9.3G	1%	/data

```

root@ip-172-31-30-211:~#

```

Reboot Instance

You can see the disk has gone

```

ubuntu@ip-172-31-30-211:~$ sudo -i
root@ip-172-31-30-211:~# df -h

```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/root	6.7G	2.1G	4.6G	31%	/
tmpfs	476M	0	476M	0%	/dev/shm
tmpfs	191M	908K	190M	1%	/run
tmpfs	476M	0	476M	0%	/tmp
none	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
none	1.0M	0	1.0M	0%	/run/credentials/systemd-resolved.service
/dev/xvda13	989M	96M	827M	11%	/boot
/dev/xvda15	105M	6.3M	99M	7%	/boot/efi
none	1.0M	0	1.0M	0%	/run/credentials/systemd-networkd.service
none	1.0M	0	1.0M	0%	/run/credentials/serial-getty@ttyS0.service
none	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	96M	8.0K	96M	1%	/run/user/1000

```

root@ip-172-31-30-211:~#

```

Make the mount persistent across reboots

Edit the fstab file:

```
vi /etc/fstab
```

Add the following line:

```
/dev/xvdh /data ext4 defaults 0 0
```

Save and exit.

```

root@ip-172-31-30-211:~# mkdir -p /data
root@ip-172-31-30-211:~# mount /dev/xvdh /data
root@ip-172-31-30-211:~# df -h

```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/root	6.7G	2.1G	4.6G	31%	/
tmpfs	476M	0	476M	0%	/dev/shm
tmpfs	191M	908K	190M	1%	/run
tmpfs	476M	0	476M	0%	/tmp
none	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
none	1.0M	0	1.0M	0%	/run/credentials/systemd-resolved.service
/dev/xvda13	989M	96M	827M	11%	/boot
/dev/xvda15	105M	6.3M	99M	7%	/boot/efi
none	1.0M	0	1.0M	0%	/run/credentials/systemd-networkd.service
none	1.0M	0	1.0M	0%	/run/credentials/serial-getty@ttyS0.service
none	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	96M	8.0K	96M	1%	/run/user/1000
/dev/xvdh	9.8G	2.1M	9.3G	1%	/data

```

root@ip-172-31-30-211:~# vi /etc/fstab

[!!] Stopped
root@ip-172-31-30-211:~# vi /etc/fstab
root@ip-172-31-30-211:~# nano /etc/fstab
root@ip-172-31-30-211:~# vi /etc/fstab
root@ip-172-31-30-211:~# cat /etc/fstab
LABEL=cloudimg-rootfs / ext4 discard,commit=30,errors=remount-ro 0 1
LABEL=BOOT /boot ext4 defaults 0 2
LABEL=UEFI /boot/efi vfat umask=0077 0 1
/dev/xvdh /data ext4 defaults 0 0
root@ip-172-31-30-211:~#

```

Reboot and verify

reboot

After reconnecting:

df -h

The /data mount should still be present.

```
ubuntu@ip-172-31-30-211:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.1G  4.6G  31% /
tmpfs            476M   0  476M   0% /dev/shm
tmpfs            191M 904K  190M   1% /run
tmpfs            476M   0  476M   0% /tmp
none            1.0M   0   1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0   1.0M   0% /run/credentials/systemd-resolved.service
/dev/xvda13      989M   96M  827M  11% /boot
/dev/xvda15      105M   6.3M   99M   7% /boot/efi
/dev/xvda1       9.8G  2.1M   9.3G   1% /data
none            1.0M   0   1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M   0   1.0M   0% /run/credentials/getty@tty1.service
none            1.0M   0   1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs           96M  8.0K   96M   1% /run/user/1000
ubuntu@ip-172-31-30-211:~$
```